

Report for Purdue’s 2026 Pontes program “New Materials for Photonic Integrated Circuits (PICs)” project

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1 Introduction

This report provides a detailed account of the research conducted within the PONTES project between January and May 2026 at the Birck Nanotechnology Center. The primary objective of this document is to systematize the theoretical framework, mathematical foundations, and technical references utilized throughout the development of this project. While this report is not structured as a formal journal article, it serves as a comprehensive technical repository intended to facilitate future reference and knowledge transfer within the research group.

2 Materials

Four distinct materials were employed in the construction of the simulated structures: ablated silicon, bare metal, air, and a silicon dioxide (SiO_2) substrate, which serves as the base for all subsequent layers. Figure 1 illustrates the physical arrangement of these materials.

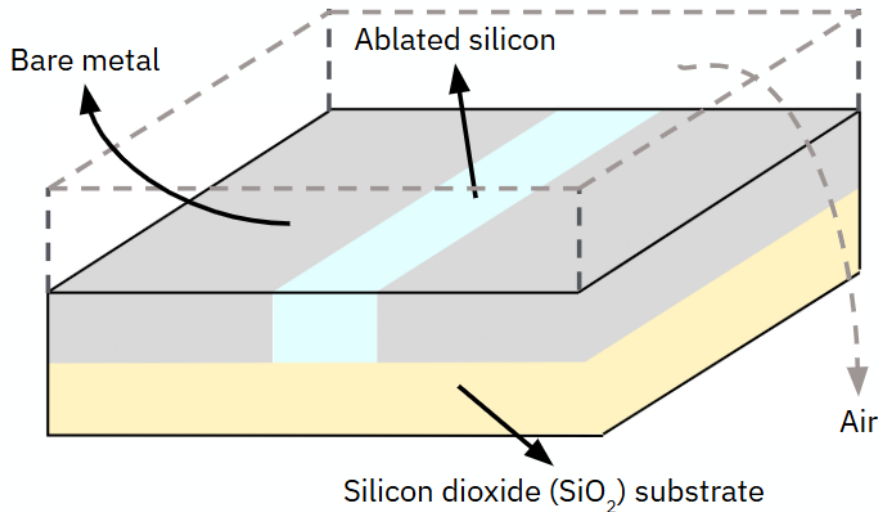


Figure 1: Materials utilized in the simulation structures.

The fabrication process involves laser-induced etching of the silicon dioxide substrate. In this process, the laser vaporizes the substrate to create micro-cavities. When additional material is

deposited onto these regions, the resulting growth characteristics differ significantly from those observed on non-etched surfaces.

Dispersion data for the ablated and bare materials specifically the permittivity as a function of wavelength, $\varepsilon(\lambda)$ were obtained from the collaborating materials research team. For air and the silicon dioxide substrate, refractive index data were sourced from the [RefractiveIndex.Info](https://refractiveindex.info) database. The refractive index for air was approximated as $n = 1$ [1], while for SiO_2 , the α -Quartz extraordinary (n_e) Sellmeier dispersion formula [2] was applied:

$$n = \sqrt{1 + \frac{0.665721\lambda^2}{\lambda^2 - 0.060^2} + \frac{0.503511\lambda^2}{\lambda^2 - 0.106^2} + \frac{0.214792\lambda^2}{\lambda^2 - 0.119^2} + \frac{0.539173\lambda^2}{\lambda^2 - 8.792^2} + \frac{1.807613\lambda^2}{\lambda^2 - 19.70^2}} \quad (1)$$

The refractive indices (RI) were derived from the complex electrical permittivities (EP) of the bare and ablated materials using the following relations [3]:

$$n = n_1 + i \cdot n_2 \quad (2)$$

where

$$(n_1)^2 = \frac{\varepsilon_1^2 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2}}{2}; (n_2)^2 = \frac{-\varepsilon_1 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2}}{2} \quad (3)$$

In these expressions, n_1 and n_2 represent the real and imaginary components of the refractive index, respectively, while ε_1 and ε_2 denote the corresponding parts of the electrical permittivity. The spectral dependence of the EP and RI for both materials is presented in Figures 2, 3, 4, and 5.

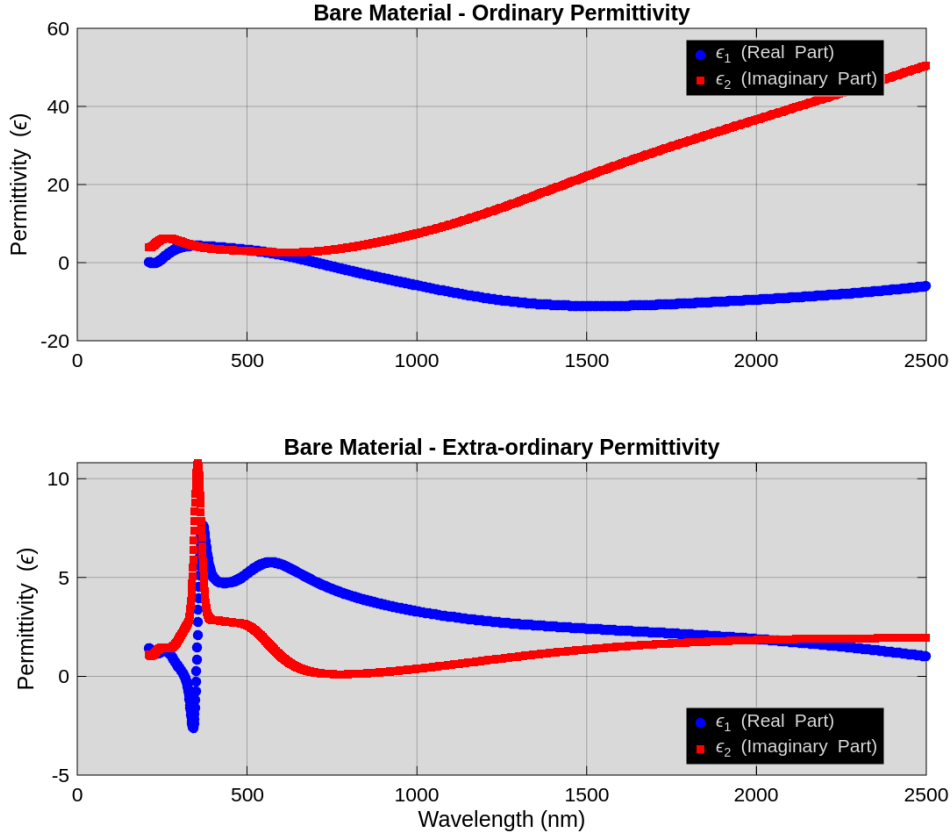


Figure 2: Electric permittivity in function of the wavelength for the bare material.

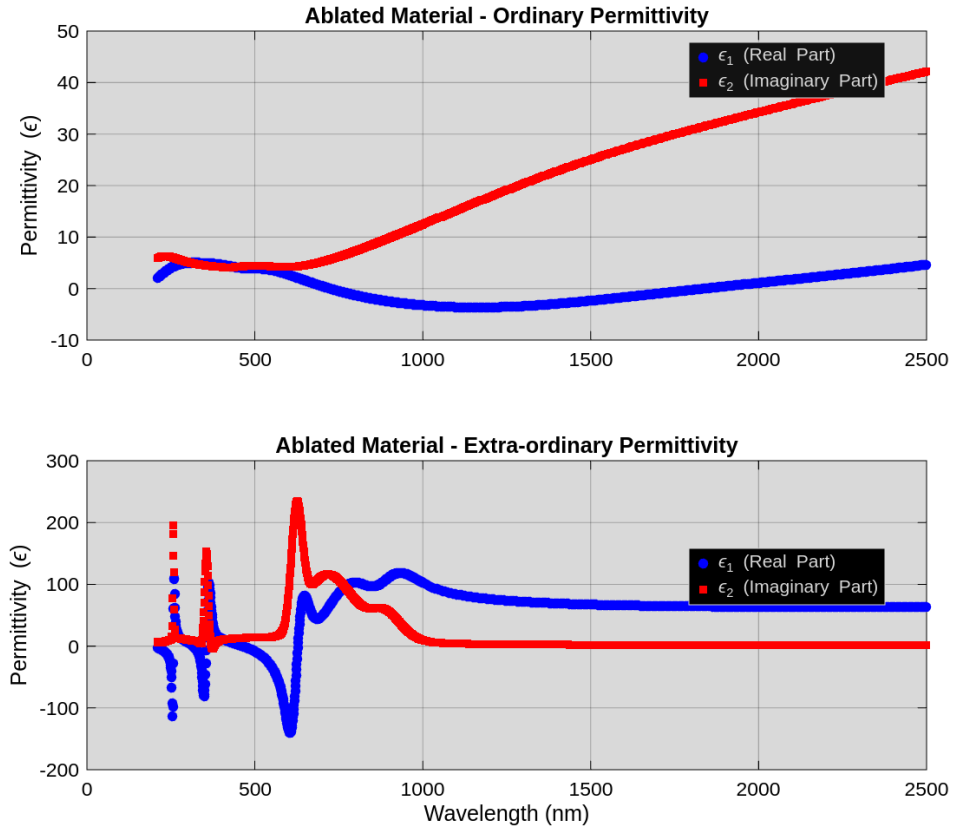


Figure 3: Electric permittivity in function of the wavelength for the ablated material.

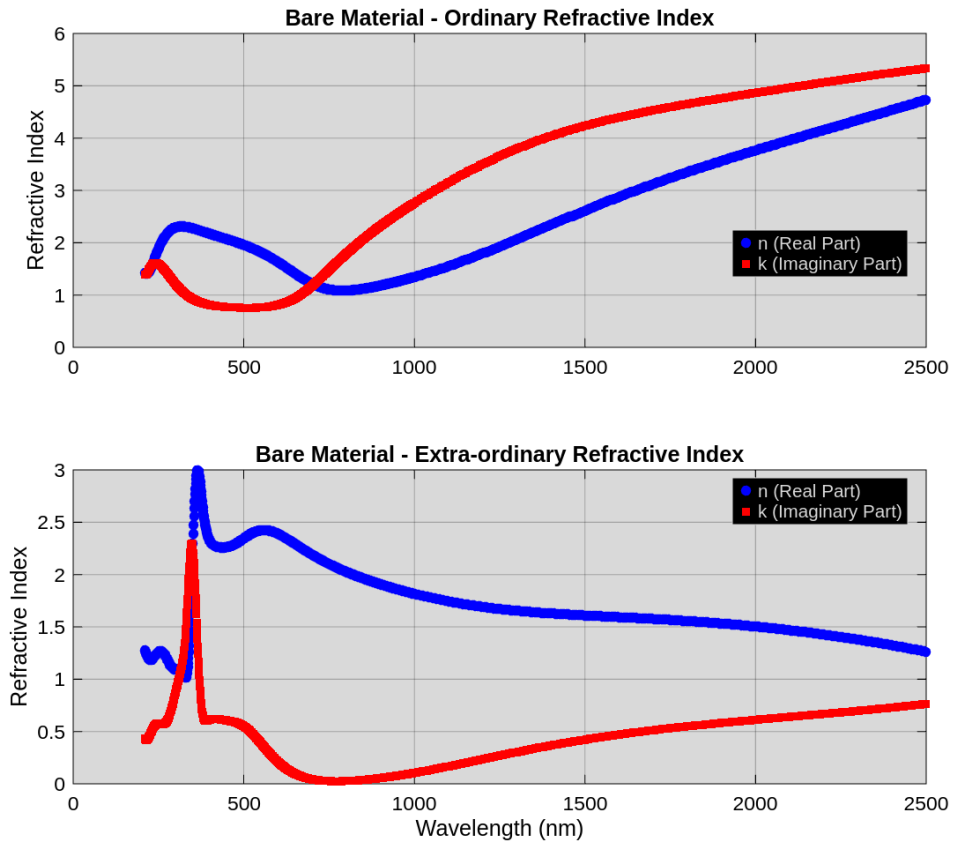


Figure 4: Refractive index in function of the wavelength for the bare material.

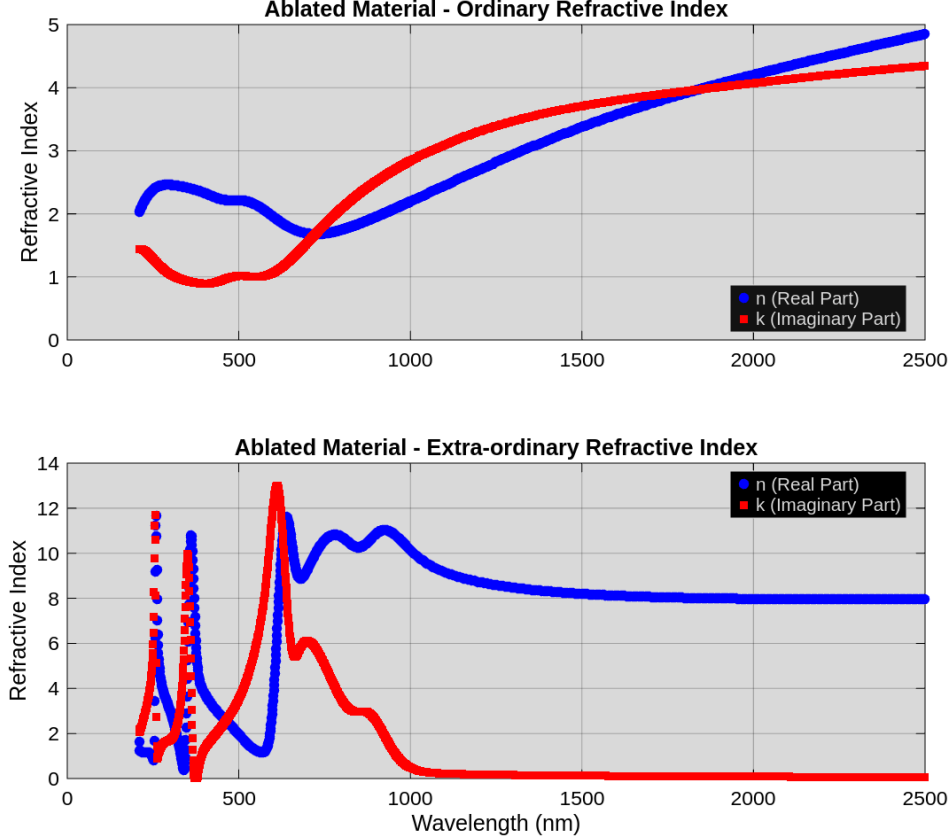


Figure 5: Refractive index in function of the wavelength for the ablated material.

Subsequent numerical experiments were conducted at a wavelength of $\lambda = 2242$ nm. This specific value was selected to ensure consistency between Beam Propagation Method (BPM) and Finite-Difference Time-Domain (FDTD/MEEP) simulations, as the latter exhibits reduced numerical stability near the 2500 nm threshold. Furthermore, at this wavelength, the imaginary component of the refractive index for the ablated material is minimized in the mid-to-far infrared spectrum (extraordinary case), resulting in lower propagation loss (see Figure 5).

The refractive indices for each material at the operational wavelength are summarized in Table 1.

Material	Refractive Index
Silica (SiO_2)	1.524
Air	1.000
Ablated Silica	$7.962 + 0.072i$
Bare Metal	$1.407 + 0.680i$

Table 1: Refractive indices at $\lambda = 2242$ nm for the materials used in the simulations.

3 Rectangular Waveguide

3.1 Structure

The initial structure investigated is a rectangular waveguide, which serves as a fundamental interconnect for light confinement and guidance between integrated photonic components.

Initial simulations using a MATLAB-based BPM implementation were performed for a waveguide with a width of $1.1413\mu\text{m}$ and a height of $1\mu\text{m}$, as illustrated in Figure 6.

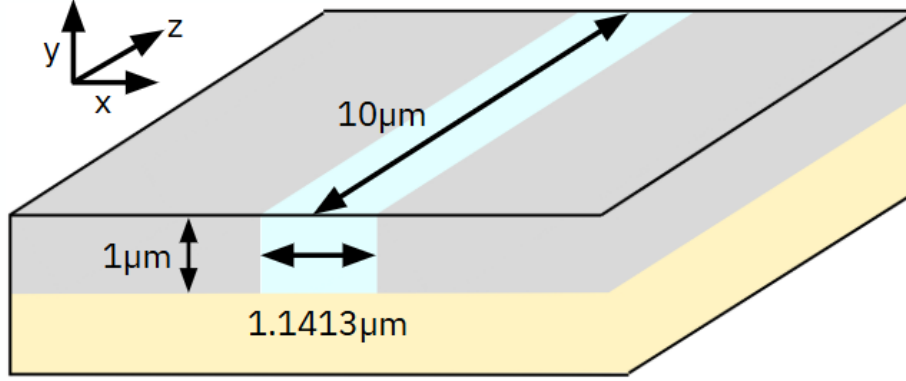


Figure 6: Schematic of the rectangular waveguide structure (not to scale).

3.2 Modes

The theoretical derivation of waveguide modes presented in this section follows established principles of electromagnetic and acoustic wave theory. Starting from the acoustic wave equation:

$$\nabla^2 p = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \quad (4)$$

Consider a hollow rectangular waveguide with a vacuum core delimited by perfectly conducting walls, as illustrated in Figure 7.

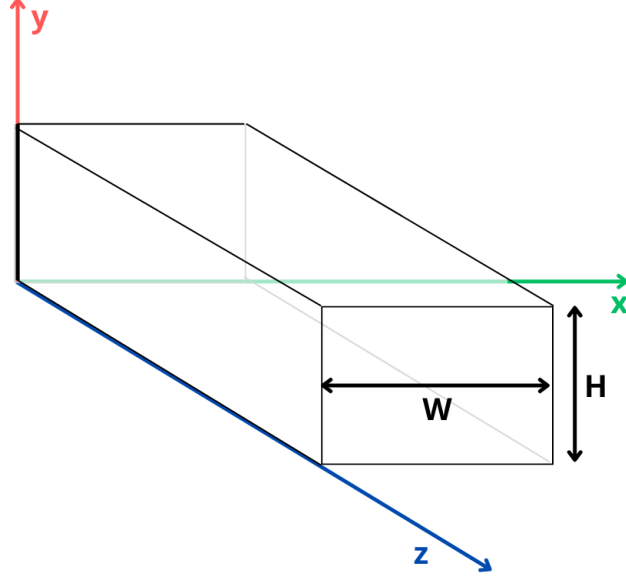


Figure 7: Rectangular waveguide geometry in Cartesian coordinates.

For a wave propagating along the z -direction, the boundary conditions require that the partial derivatives of the pressure p vanish at the boundaries: $\frac{\partial p}{\partial y} = 0$ at $y = 0$ and $y = H$, and $\frac{\partial p}{\partial x} = 0$ at $x = 0$ and $x = W$. This confinement is analogous to a vibrating string fixed at its extremities, where the field amplitude must satisfy the constraints imposed by the waveguide walls.

Expanding Equation 4 into its directional components yields:

$$\nabla^2 p = \frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} \quad (5)$$

Applying the method of separation of variables, a solution of the form $p(x, y, z, t) = X(x)Y(y)Z(z)T(t)$ is assumed. Using partial derivative notation (e.g., $X_{xx} = \frac{\partial^2 X}{\partial x^2}$), the wave equation is rewritten as:

$$X_{xx}YZT + XY_{yy}ZT + XYZ_{zz}T = \frac{1}{c^2}XYZT_{tt} \quad (6)$$

Dividing the expression by $p = X(x)Y(y)Z(z)T(t)$ results in:

$$\frac{X_{xx}}{X} + \frac{Y_{yy}}{Y} + \frac{Z_{zz}}{Z} = \frac{1}{c^2} \frac{T_{tt}}{T} \quad (7)$$

Since each term must equal a constant, the separation constants are defined as $\frac{X_{xx}}{X} = -k_x^2$, $\frac{Y_{yy}}{Y} = -k_y^2$, and $\frac{Z_{zz}}{Z} = -k_z^2$. The wavevector k is then defined by the relation:

$$k^2 = k_x^2 + k_y^2 + k_z^2 \quad (8)$$

leading to the temporal characteristic equation:

$$-k^2 = \frac{1}{c^2} \frac{T_{tt}}{T} \quad (9)$$

These relations describe simple harmonic motion, supporting solutions in the form of sinusoids or complex exponentials. The solutions for the x and y directions represent standing waves: $X(x) = A_x \cos(k_x x)$ and $Y(y) = A_y \cos(k_y y)$, where A_x and A_y are amplitudes, and the transverse wavenumbers are quantized as $k_x = \frac{m\pi}{W}$ and $k_y = \frac{n\pi}{H}$, with $m, n \in \mathbb{Z}$.

The combined solution for the z -direction and time t is expressed as $Z(z)T(t) = A_{zt} \cos(\omega t - k_z z + \phi)$. Consequently, the complete expression for the pressure field p is:

$$p = A \cos(k_x x) \cos(k_y y) \cos(\omega t - k_z z + \phi) \quad (10)$$

where A is the total amplitude, $k^2 = k_x^2 + k_y^2 + k_z^2$, and $k = \frac{\omega}{c}$.

Modes are defined as the discrete sets of solutions corresponding to the integer indices m and n . Physically, m and n represent the number of field extrema (maxima and minima) along the x and y axes, respectively.

Mode dispersion occurs when a specific mode can no longer be supported by the waveguide geometry. Isolating the longitudinal wavenumber k_z from Equation 8 gives:

$$k_z = \sqrt{k^2 - k_x^2 - k_y^2} = \sqrt{\frac{\omega^2}{c^2} - \frac{m^2 \pi^2}{W^2} - \frac{n^2 \pi^2}{H^2}} \quad (11)$$

Propagating solutions require k_z to be a real number, which occurs only if:

$$\frac{\omega^2}{c^2} \geq \frac{m^2 \pi^2}{W^2} + \frac{n^2 \pi^2}{H^2} \quad (12)$$

The physical interpretation of the propagation constant k_z results in three distinct cases:

1. If $k_z^2 < 0$, the mode is evanescent and decays exponentially as it attempts to propagate.
2. If $k_z = 0$, the system is at the cut-off frequency, forming a standing wave with no net energy flux along the z -axis.
3. If $k_z^2 > 0$, the mode propagates through the waveguide, albeit at a lower phase velocity compared to lower-order modes.

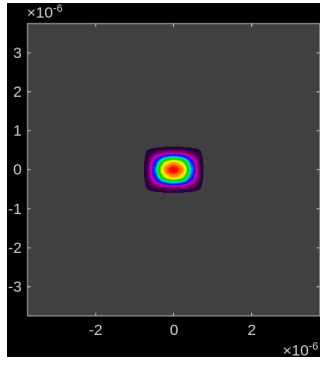
Representative mode profiles generated by the MATLAB-based BPM solver for the studied rectangular waveguide are presented in Figure 8. While the analytical derivation assumes an ideal hollow waveguide, these numerical results account for the specific material properties and geometry of the simulated structure. It is important to note that the $TE_{0,0}$ mode is not physically realizable in rectangular waveguides; therefore, the analysis focuses on modes starting from the $TE_{1,0}$ fundamental order.

3.3 Data Analysis and Results

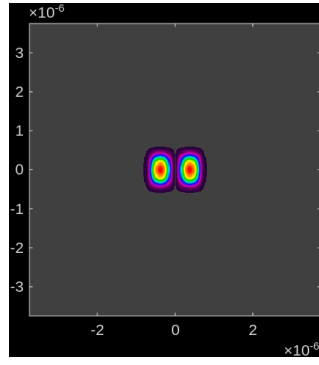
A simulation was developed for the standard rectangular waveguide geometry previously depicted in Figure 7. The analysis utilized the fundamental (first-order) mode at a wavelength of $\lambda = 2242$ nm, corresponding to the "Mode 1" profile in Figure 8. The primary metric for evaluating optical loss is the electric field power, defined as:

$$P = \sum \frac{1}{2} c \varepsilon_0 |E|^2 \Delta x \Delta y \quad (13)$$

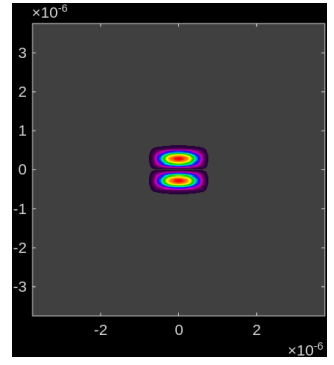
where c is the speed of light, ε_0 is the permittivity of free space, $|E|$ represents the electric field intensity, and $\Delta x, \Delta y$ denote the pixel dimensions.



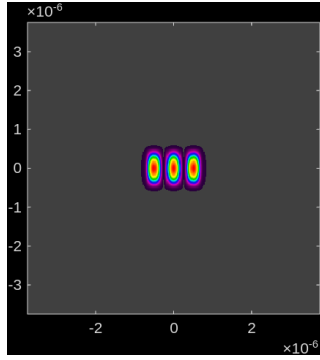
(a) Mode 1



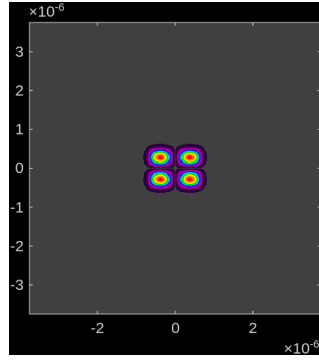
(b) Mode 2



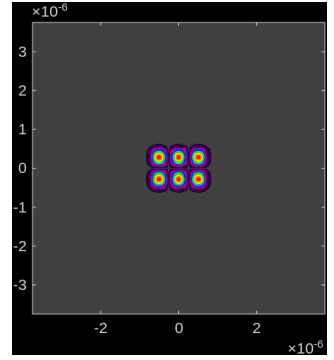
(c) Mode 3



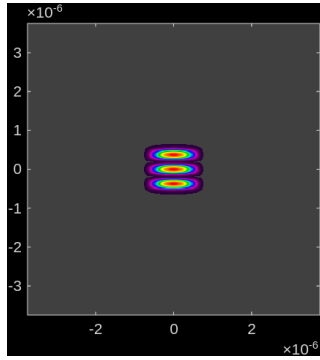
(d) Mode 4



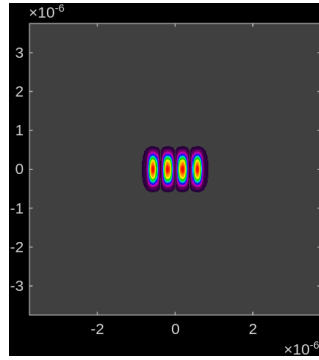
(e) Mode 5



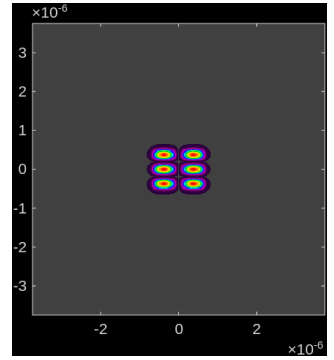
(f) Mode 6



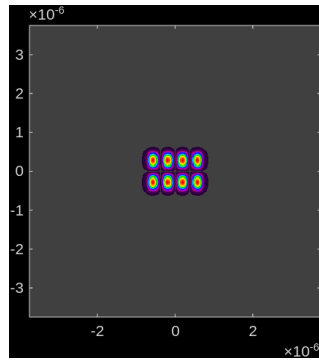
(g) Mode 7



(h) Mode 8



(i) Mode 9



(j) Mode 10

Figure 8: TE-mode profiles identified by the BPM-MATLAB solver for a rectangular waveguide.

By employing relative power measurements (e.g., remaining guided power), the constant coefficients are eliminated, resulting in an expression based solely on the electric field intensity:

$$\frac{P_2(x, y)}{P_1(x, y)} = \frac{\sum_{y=Y_1}^{Y_2} \sum_{x=X_1}^{X_2} |E_2(x, y)|^2}{\sum_{y=Y_1}^{Y_2} \sum_{x=X_1}^{X_2} |E_1(x, y)|^2} \times 100 \quad (14)$$

where X_1, X_2 and Y_1, Y_2 represent the horizontal and vertical integration limits (in pixels), respectively, for the region of interest. The result is expressed as a percentage of the reference power.

The simulation results are summarized in Table 2, which details the definitions of E_1 and E_2 , the calculated relative power percentages, and their physical interpretations. Corresponding data for a lossless scenario (where the imaginary part of the refractive index is set to $n_2 = 0$) are provided in Table 3.

Metric No.	E_2	E_1	P_2/P_1 [%]	Interpretation
1	Initial E-field (core)	Initial total E-field	99.82	Initial mode confinement and coupling efficiency.
2	Final total E-field	Initial total E-field	1.68	Total remaining power (guided and unguided).
3	Final E-field (core)	Initial E-field (core)	1.67	Remaining guided power.
4	Final E-field (core)	Final total E-field	99.62	Fraction of field intensity retained within the core.

Table 2: Relative power metrics for the waveguide considering absorption in the ablated regions.

Metric No.	E_2	E_1	P_2/P_1 [%]	Interpretation
1	Initial E-field (core)	Initial total E-field	99.82	Initial mode confinement and coupling efficiency.
2	Final total E-field	Initial total E-field	96.18	Total remaining power (guided and unguided).
3	Final E-field (core)	Initial E-field (core)	96.16	Remaining guided power.
4	Final E-field (core)	Final total E-field	99.80	Fraction of field intensity retained within the core.

Table 3: Relative power metrics for the waveguide assuming zero absorption ($n_2 = 0$) in the ablated regions.

The numerical results indicate that the electromagnetic field is effectively confined within the core, as evidenced by the high values of Metric 4 in both cases. Furthermore, Metric 1 demonstrates efficient mode coupling at the device input. However, a comparison between the two tables reveals that Metrics 2 and 3 are significantly lower in the absorptive case, indicating substantial attenuation.

The guided power loss, expressed in dB/ μm , was calculated using the standard power ratio formula $10\log_{10}(P_o/P_i)$, normalized by the $10\mu\text{m}$ length of the device. For the case including ablated material absorption, the propagation loss was determined to be $-1.777\text{ dB}/\mu\text{m}$. In contrast, the lossless material simulation yielded a significantly lower value of $-0.017\text{ dB}/\mu\text{m}$. These findings suggest that the predominant loss mechanism is material absorption.

As illustrated in Figure 5, the extraordinary imaginary coefficient for the material deposited on ablated silica at $\lambda = 2242\text{ nm}$ is among the lowest within the analyzed spectral range. Consequently, the observed performance is primarily limited by the intrinsic properties of the material at the specified operational wavelength.

4 Mach-Zehnder Interferometer

4.1 Structure

A Mach-Zehnder Interferometer (MZI) is an electro-optic device comprising a power splitter and a combiner. Typically, the application of an external electric field induces a change in the refractive index (RI) of one of the interferometer arms. This modulation alters the propagation velocity within that arm, resulting in a phase difference between the two electromagnetic fields. Upon recombination at the output, constructive or destructive interference occurs depending on the phase shift. A longitudinal schematic of the simulated MZI is presented in Figure 9, where R denotes the radius of curvature and d represents the maximum separation between the centers of the arms.

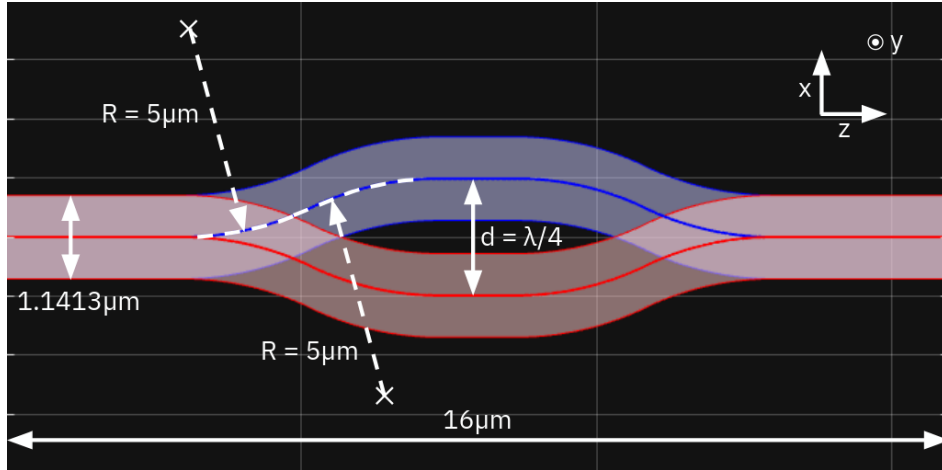


Figure 9: Longitudinal view of the Mach-Zehnder Interferometer structure.

The structure was implemented in the BPM-MATLAB environment using longitudinal cross-sections. The geometry of each arm is defined by functions describing the transverse position (x) as a function of the propagation distance (z). Due to the symmetry of the device, the path of the upper arm (depicted in blue in Figure 9) is derived first, and the lower arm (red) is obtained by a sign inversion of the x -coordinates.

The initial section of the splitter consists of a circular arc extending from $\theta = \pi$ (where $x = 0$)

to a specific angle where $x = d/2$. This trajectory is described by:

$$x = R \left(1 - \cos \left(\arcsin \left(\frac{(z-l)}{R} \right) \right) \right) \quad (15)$$

where l represents the length of the initial straight segment and R is the radius of curvature.

The second segment of the splitter originates at $x = d/2$ and concludes in a straight waveguide section. To ensure tangential continuity, the center of the second circular arc must be positioned at:

$$(z, x) = \left(l + 2\sqrt{\frac{dR}{2} - \frac{d^2}{16}}, \frac{d}{2} - 2R \right) \quad (16)$$

This coordinate is derived from the requirement that the two circumferences, centered at C_1 and C_2 , share a common tangent point P . This point is located at $P(z, x) = \left(\sqrt{\frac{dR}{2} - \frac{d^2}{16}}, d/4 \right)$, calculated by solving Equation 15 for $x = d/4$. Since both arcs have an identical radius R , P serves as the midpoint between C_1 and C_2 .

To complete the path definition, the arc orientation is adjusted to the interval $[0, \pi]$. The resulting expression for the second arc is given by Equation 17:

$$x = \left(\frac{d}{2} - 2R \right) + R \left(1 - \cos \left(\arcsin \left(\frac{z - \left(l + 2\sqrt{\frac{dR}{2} - \frac{d^2}{16}} \right)}{R} \right) - \pi \right) \right) \quad (17)$$

The geometric model can be validated through numerical plotting tools, such as the [Desmos calculator](#), by substituting the spatial variables (z, x) with Cartesian coordinates (x, y) .

For the combiner section, the curves are defined as symmetrical counterparts to the splitter trajectory, displaced along the transverse axis. The first and second segments of the combiner are represented by Equations 18 and 19, respectively:

$$x = \frac{d}{2} - \left(R \left(1 - \cos \left(\arcsin \left(\frac{(z-l)}{R} \right) \right) \right) \right) \quad (18)$$

$$x = \frac{d}{2} - \left(\left(\frac{d}{2} - 2R \right) + R \left(1 - \cos \left(\arcsin \left(\frac{z - \left(l + 2\sqrt{\frac{dR}{2} - \frac{d^2}{16}} \right)}{R} \right) - \pi \right) \right) \right) \quad (19)$$

In the simulation environment, the longitudinal coordinate z is reset to zero at the start of each new segment. Consequently, the parameter l may be set to zero for modular path calculation within the software.

4.2 Destructive Interference

As previously established, the Mach-Zehnder Interferometer (MZI) operates by inducing a phase shift in one of its arms to facilitate interference upon recombination at the output. When the fields from both arms are recombined at the combiner, a phase difference of π results in destructive interference, effectively canceling the output signal. The refractive index modulation required to achieve this condition is derived from established Photonic Integrated Circuit (PIC) theory.

The optical power at the output of the interferometer is described by Equation 20, where $\Delta\phi$ represents the cumulative phase difference between the waves propagating in each arm, as defined by Equation 21:

$$P_{out} = P_{in} \cos^2 \left(\frac{\Delta\phi}{2} \right) \quad (20)$$

$$\Delta\phi = \int_a^b k_0 n_{eff}(l) dl - \int_a^b k_0 n'_{eff}(l) dl \quad (21)$$

In these expressions, $n_{eff}(l)$ and $n'_{eff}(l)$ are the effective refractive indices as functions of the segment length, a and b denote the initial and final points of the modulation region, and $k_0 = 2\pi/\lambda_0$ is the free-space wavenumber.

Assuming that the refractive index modulation is uniform and restricted to the straight section between the splitter and the combiner, the phase difference simplifies to:

$$\Delta\phi = k_0 L \Delta n_{eff} \quad (22)$$

The effective refractive index for the fundamental mode ($n_{eff} = 7.843$) was determined numerically using the `findModes` function in BPM-MATLAB. To achieve a phase shift via mechanisms such as carrier injection or thermo-optic effects, the relationship between the change in the effective index (Δn_{eff}) and the change in the core refractive index (Δn_{core}) must be established.

This relationship is defined by $\Delta n_{eff} = \Gamma \Delta n_{core}$, where Γ represents the modal confinement factor. Given that Mode 1 is a strongly guided mode—with a confinement factor of 99.82% as indicated by Metric 1 in Tables 2 and 3—it is reasonable to approximate $\Gamma \approx 1$. Consequently, $\Delta n_{eff} \approx \Delta n_{core}$.

Considering a straight section length of $L = 1.337 \mu\text{m}$ (calculated by subtracting the bend sections and initial straight segments from the total device length), the condition for total destructive interference ($\Delta\phi = \pi$) is applied to Equation 22:

$$\pi = k_0 \cdot 1.337 \mu\text{m} \cdot \Delta n_{eff} \Rightarrow \Delta n_{eff} = \frac{\lambda_0}{2 \cdot 1.337 \mu\text{m}} = \frac{2242 \text{ nm}}{2 \cdot 1.337 \mu\text{m}} = \mathbf{0.8384} \quad (23)$$

Therefore, a refractive index increase of 0.8384 in one of the interferometer arms is required to achieve complete destructive interference and minimize the output power.

4.3 Data Analysis and Results

The data analysis for the Mach-Zehnder Interferometer (MZI) is based on the ratio between the output and input optical power. The localized modification of the refractive index is illustrated in Figure 10, where the index of the straight section in the upper arm is manually adjusted.

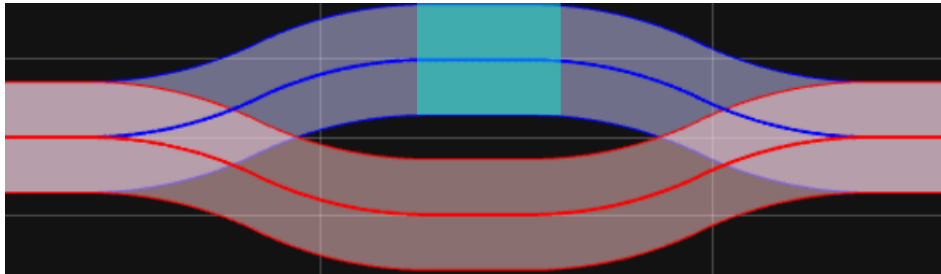


Figure 10: Schematic of the MZI indicating the region of refractive index modulation (highlighted in blue; not to scale).

The primary performance metric evaluated is the remaining guided power. Table 4 presents the power data for the reference case ($\Delta n_{core} = 0$), while Table 5 details the relative powers for the modulated case ($\Delta n_{core} = 0.8384$).

Metric No.	E_2	E_1	P_2/P_1 [%]	Interpretation
1	Initial E-field (core)	Initial total E-field	99.81	Initial mode confinement and coupling efficiency.
2	Final total E-field	Initial total E-field	19.58	Total remaining power (guided and unguided).
3	Final E-field (core)	Initial E-field (core)	18.90	Remaining guided power.
4	Final E-field (core)	Final total E-field	96.37	Fraction of field intensity retained within the core.

Table 4: Power ratios for the MZI simulation with $\Delta n_{core} = 0$.

Metric No.	E_2	E_1	P_2/P_1 [%]	Interpretation
1	Initial E-field (core)	Initial total E-field	99.81	Initial mode confinement and coupling efficiency.
2	Final total E-field	Initial total E-field	19.95	Total remaining power (guided and unguided).
3	Final E-field (core)	Initial E-field (core)	18.96	Remaining guided power.
4	Final E-field (core)	Final total E-field	94.83	Fraction of field intensity retained within the core.

Table 5: Power ratios for the MZI simulation with $\Delta n_{core} = 0.8384$.

The numerical results indicate that the MZI does not exhibit the expected ideal behavior. Paradoxically, the guided power in the case where destructive interference should be dominant ($\Delta n_{core} = 0.8384$) is marginally higher than in the reference case. Furthermore, the output power for the $\Delta n_{core} = 0$ scenario is considerably lower than anticipated. An investigation of the electric field distributions is necessary to identify the causes of this deviation.

The cross-sectional phase profiles of the electric fields after the straight sections (prior to recombination) for both cases are shown in Figure 11.

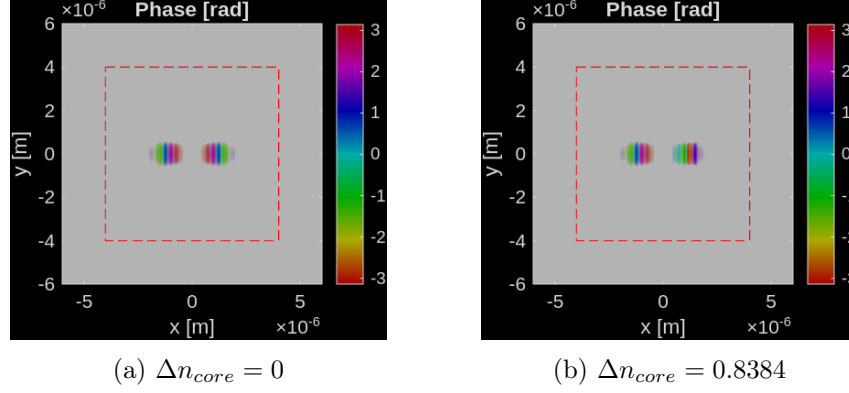


Figure 11: Electric field phase distribution after the straight modulation section for both refractive index configurations.

In the modulated case (Figure 11b), the symmetric vertical phase fronts exhibit a phase difference of approximately π . This confirms that the calculated Δn_{core} successfully induces the target phase shift. However, despite this correct phase relationship, effective intensity cancellation is not observed upon recombination.

The electric field profiles following the arm recombination (before the final straight waveguide section) are presented in Figure 12. Corresponding intensity plots along the transverse axis (indicated by the horizontal green lines) are provided in Figure 13.

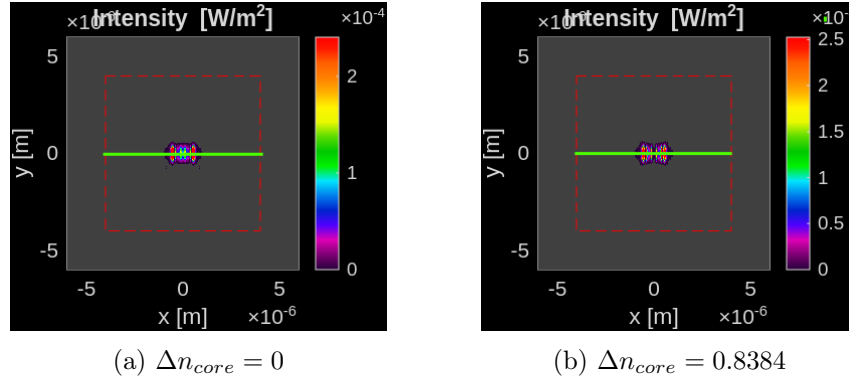


Figure 12: Electric field intensity profiles at the junction of the MZI arms.

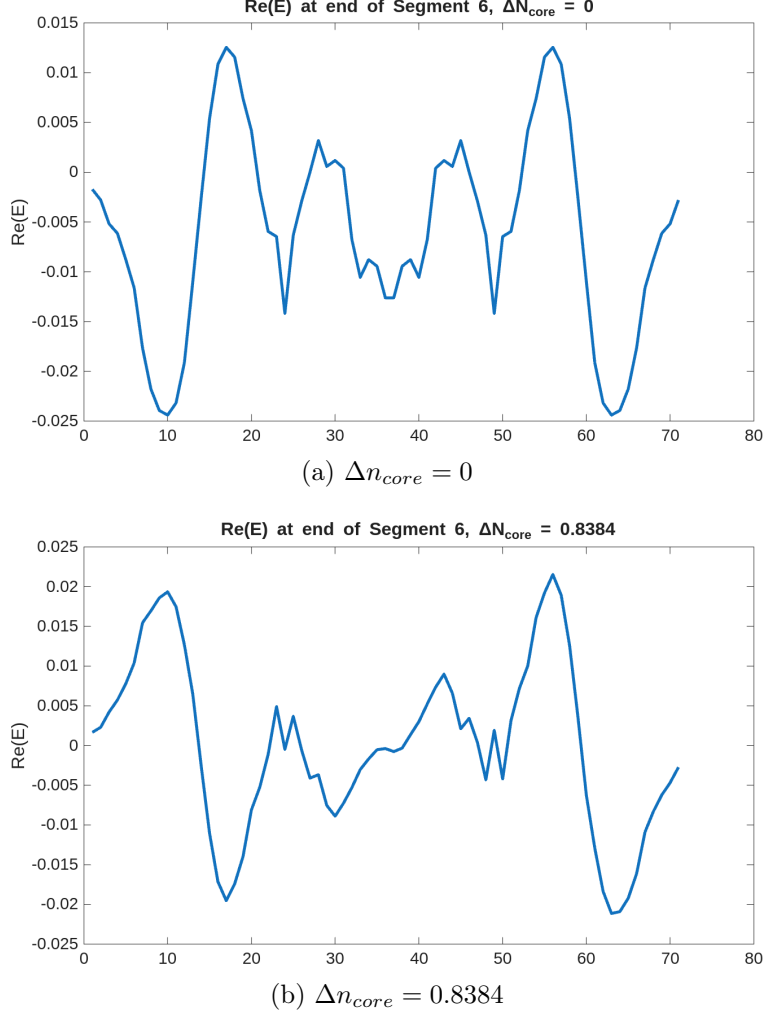


Figure 13: Transverse electric field intensity distributions for the non-modulated and modulated scenarios.

4.4 Proposed Optimizations

To improve the performance and functional characteristics of the interferometer, three consecutive structural modifications were implemented and evaluated.

The first optimization targeted the enhancement of the destructive interference effect. As indicated by the field distributions in Figure 13, the reference case ($\Delta n_{\text{core}} = 0$) exhibits a symmetric profile, whereas the modulated case ($\Delta n_{\text{core}} = 0.8384$) is characterized by an antisymmetric profile. Theoretically, the introduction of a funnel-like tapering structure facilitates the interaction between these field distributions, enabling more effective intensity cancellation. A tapered waveguide with an input width of $1.1413 \mu\text{m}$, an output width of $0.3 \mu\text{m}$, and a length of $20 \mu\text{m}$ was integrated between the combiner and the device output. The lengths of the straight arm sections remained unchanged. The resulting guided power values are summarized in Table 6.

Δn_{core}	Remaining Guided Power [%]
0	1.36
0.8384	0.15

Table 6: Output guided power for the $20\ \mu\text{m}$ taper optimization under different modulation settings.

While the integration of the taper allows the MZI to function more effectively as an optical switch, the output power in the non-modulated state remains notably low. Consequently, a second optimization was implemented to mitigate losses attributed to back-reflection. This involved increasing the radius of curvature (R) in both the splitter and combiner sections. A numerical iteration was performed for radii ranging from $5\ \mu\text{m}$ to $50\ \mu\text{m}$ in steps of $0.5\ \mu\text{m}$. The dependence of the MZI output power on R for the $\Delta n_{core} = 0$ case is illustrated in Figure 14.

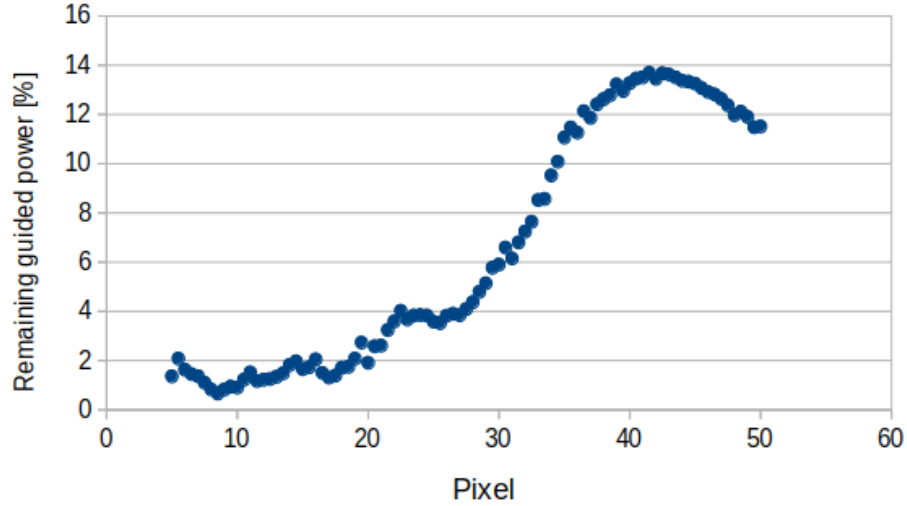


Figure 14: Output power as a function of the radius of curvature (R) for the $\Delta n_{core} = 0$ configuration.

Within the simulated range, a maximum output power of 13.69% was achieved at $R = 41.5\ \mu\text{m}$, representing a 10.07-fold increase compared to the initial taper-integrated configuration. As shown in Table 7, the destructive interference remains effective under these conditions.

Δn_{core}	Remaining Guided Power [%]
0	13.69
0.8384	0.58

Table 7: Output guided power for the combined taper and optimized radius configuration.

A third structural modification was evaluated to investigate whether a longer central modulation section could mitigate potential back-reflection at the interface of the modified refractive index, where

the optical signal transitions between media of different refractive indices. However, as illustrated in the transverse field profiles in Figure 13(b), the intensities emerging from both arms remain comparable. Consequently, no significant evidence of back-reflection was observed in the modulated arm. The final, optimized MZI architecture is presented in Figure 15.

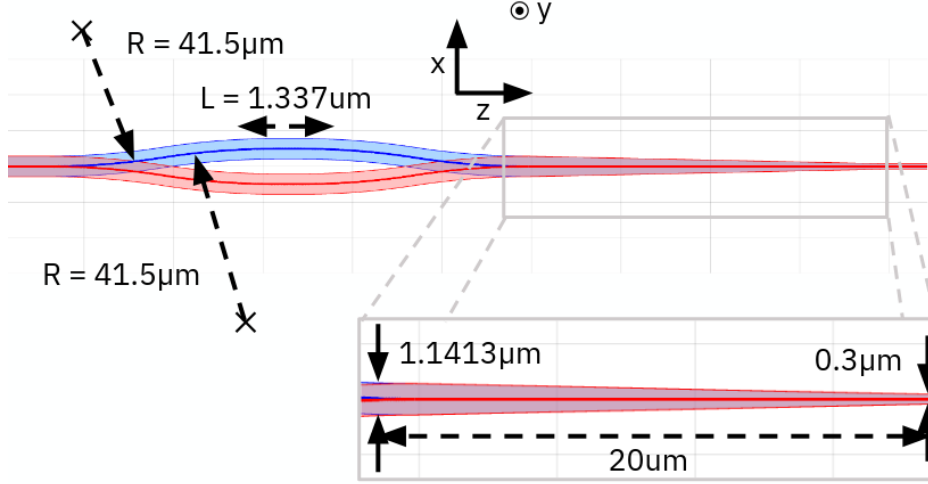


Figure 15: Top-view of the Mach-Zehnder Interferometer structure after the three proposed optimizations.

When no refractive index modulation is applied ($\Delta n_{core} = 0$), the structure achieves a maximum output power of 72.43% in relation to the maximum output power of the initial structure. Upon the induction of the target refractive index difference, the output power decreases to 0.58%. This corresponds to a 96.94% reduction in transmitted power compared to the initial non-optimized structure, demonstrating a high extinction ratio and significantly improved modulation performance.

5 Scripts

This section correlates the technical descriptions provided in previous chapters with their respective MATLAB implementations. Detailed documentation and internal comments are provided within the source code of each script to facilitate technical review and further development.

The computational requirements for executing these scripts include MATLAB (version R2025a or later) and the BPM-MATLAB simulation framework, which is accessible via GitHub [here](#). Additionally, the spectral dispersion data for the ablated and bare materials must be located within the `materials_data` directory. The `BPM-Matlab` folder must be included in the MATLAB path prior to execution.

5.1 Materials Section

The spectral dependence plots for permittivity and refractive index (Figures 2, 3, 4, and 5) were generated using the `PlotEPandRI.m` script. This utility produces interactive visualizations, allowing for localized data inspection and the selection of specific material properties (RI or EP) for analysis.

5.2 Rectangular Waveguide Section

The numerical simulations and corresponding data analysis for the fundamental waveguide structure are contained within `Waveguide.m`.

5.3 Mach-Zehnder Interferometer Section

The analysis of the Mach-Zehnder Interferometer (MZI) is supported by four primary scripts. The longitudinal schematic of the device (Figure 9) was produced by `MZI_TopView.m`. The baseline simulation and electric field power acquisition are implemented in `MZI_InitialVersion.m`. Numerical iterations for the optimization of the radius of curvature (as shown in Figure 14) are performed by `MZI_RadiusIteration.m`. Finally, the fully optimized MZI configuration, incorporating the tapered waveguide and the extended modulation section, is implemented in `MZI_Optimized.m`.

5.4 Auxiliary Visualization Tools

The script `LongitudinalPlot.m` enables the visualization of longitudinal electric field cross-sections along a specified xz plane. The tool provides three distinct visualization modes: phase (in radians), real component ($Re(E)$), and field intensity ($|E|^2$). This utility facilitates the inspection of field propagation behavior throughout the photonic structure.

To utilize this tool, the simulation parameter `P.storeE3D` must be set to `true`. Furthermore, the MATLAB workspace must remain intact following simulation completion, as the visualization script directly references active variables within the environment.

Currently under development, `PhotonicCircuitBlocks.m` is a wrapper framework designed to abstract the complex syntax and structural implementation details of the BPM-MATLAB environment into simplified, high-level commands (e.g., `addWaveguide()`). This project aims to streamline the modeling process through a modular and scalable approach to photonic circuit design.

6 Conclusion and Future Work

The structural optimizations implemented in this study successfully addressed the initial design objectives, particularly in facilitating functional destructive interference within the Mach-Zehnder Interferometer. Although the transmission efficiency in the "on" state exhibited a relative decrease during the iterative refinement, the induction of a refractive index differential proved effective in achieving a clear modulation response. A primary constraint of the current analysis is the reliance on a lossless model for the ablated material; given its significant intrinsic absorption, future iterations must integrate these material losses to provide a more comprehensive performance profile beyond purely geometric optimizations.

Future research will focus on the cross-validation of these results through alternate numerical methods, specifically the Finite-Difference Time-Domain (FDTD) method. Furthermore, a broader parameter space will be explored, including multi-variable iterations of the modulation segment lengths and the output dimensions of the tapered structures. Collaboration with materials science teams is also anticipated to investigate the fabrication of low-loss alternatives within the targeted optical spectrum.

The proposed architecture demonstrates significant potential for the development of high-speed photonic switches. Its scalability suggests a pathway toward complex integrated components, such as optical logic gates, which are foundational for the advancement of next-generation photonic computational architectures.

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